

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 3.2: Current Electricity — Teacher Reference (Pre-filled)

SUMMARY TABLE: PHYSICS GRADE 10	
Sub-Strand	Sub-Strand 3.2: Current Electricity
Driving Question	DRIVING QUESTION: Why do the other tubes stay lit when one blows out, and why do bulbs connected in series glow dimmer than identical bulbs connected in parallel to the same battery?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Blown Tube and the Dim Bulbs: Anchoring Phenomenon	Describe what you saw: how bright the three identical bulbs were in series and in parallel on the same 9 V battery pack, what happened when one bulb was removed in each circuit, and what happened when one of the four tubes in the Nairobi classroom blew out and when a bulb in an old Christmas-light string failed.	Write your first ideas about why the connection pattern changes brightness and why the other tubes stay lit. Say what you are unsure about, and note that these are initial ideas which you will test with evidence.	Write one 'I wonder...' question for the Driving Question Board and say what you would test next with the 9 V pack and bulbs. Do not give a final explanation yet, because the full explanation will be built in later lessons.
Lesson 2 Circuit Components and Symbols: What Does a Bulb Need to Light?	We observed that the bulb lit only when the 9 V pack, wires and bulb formed a complete closed loop. When we opened the switch, removed a wire or left a gap, the bulb went out. In the diagrams of the demo, the three series bulbs share one loop, whereas the parallel bulbs are on separate branches across the same 9 V pack.	We learned that a circuit needs a source and an unbroken path to work, and that standard symbols let us draw circuit diagrams that anyone can read and build. 'In series' means one loop and 'in parallel' means branches. A break in a single loop gives an open circuit, so everything in that loop stops.	The Christmas-light string goes dark when one bulb fails because all the bulbs are in one series loop, so one break opens the loop. In the Nairobi classroom, each tube seems to have its own branch, so the loop for each of the other tubes is still complete when one blows. We still need evidence to explain why series bulbs are dim and parallel bulbs are bright.
Lesson 3 Electric Current: Charge Flow and the Ammeter in Series	We measured the current in the series loop of three identical 9 V, 2.7 W bulbs on a 9 V battery pack: 0.10 A at every point, and 0 A (all bulbs out) when one bulb was removed. In the parallel circuit we measured 0.30 A in each branch and 0.90 A at the battery, and the bulbs were brighter.	Current is the flow of charge, $I = Q/t$, in amperes (1 A = 1 C/s). An ammeter is connected in series. The current is the same at every point in a series loop, so bulbs do not use up current. Conventional current flows from + to -, while electrons move the opposite way. In 60 s a current of	So far, a series loop has a single path so the same 0.10 A passes every point, and a broken bulb opens the loop, which is why an old Christmas-light string goes dark. We cannot yet explain why the series bulbs are dim, why the parallel battery gives 0.90 A, or why the other Nairobi tubes stay lit.

		0.10 A moves 6 C.	These need potential difference and resistance, which we study next.
Lesson 4 Potential Difference and e.m.f.: Energy Per Charge in the Dim and Bright Bulbs	We measured 1.5 V across one dry cell, 4.5 V across three cells in series and 9.0 V across the whole battery. In the series circuit, each of the three identical bulbs had 3.0 V across it ($3.0 + 3.0 + 3.0 = 9.0$ V) and glowed dim. In the parallel circuit each bulb had 9.0 V across it and glowed bright. When a series bulb was removed, all the lights went out.	Potential difference (p.d.) is the energy transferred per unit charge between two points, measured in volts ($1 \text{ V} = 1 \text{ J/C}$) with a voltmeter connected in parallel across the component. e.m.f. is the energy per unit charge supplied by the source, such as the 9 V battery. A series bulb receives only part of the e.m.f. (3.0 V), while a parallel bulb receives all of it (9.0 V).	The parallel bulbs glow brighter because each coulomb of charge gives up all 9 J in one bulb, while in series the 9 J is divided between three bulbs, 3 J each. This helps explain the dim series bulbs and suggests that each tube in the Nairobi classroom is across the full supply. We still need to explain why the p.d. is shared equally and why the parallel circuit draws more current, which needs resistance and Ohm's law.
Lesson 5 Ohm's Law: How Much Current Does a Resistor Allow for Each Volt?	We measured the p.d. across and the current through a 30Ω resistor at 1.5 V, 3.0 V, 4.5 V, 6.0 V and 7.5 V and found currents of 0.05 A, 0.10 A, 0.15 A, 0.20 A and 0.25 A. The graph of V against I was a straight line through the origin, and V/I stayed at 30Ω . The 30Ω bulb at 3.0 V carried 0.10 A.	For an ohmic conductor at constant temperature, the current is directly proportional to the p.d. ($V = IR$). Resistance is found from V/I or from the gradient of the V-I graph (30Ω here). A larger p.d. across the same resistance gives a larger current, and a larger resistance gives a smaller current for the same p.d.	This gives us a rule to explain the dim series bulbs and the bright parallel bulbs: a parallel bulb has 9 V across 30Ω , so $I = 9/30 = 0.30$ A. We still need to find out how the three bulbs in series combine to give 90Ω and 0.10 A, and why a failed tube in the Nairobi classroom does not stop the others. This will come in the lessons on resistance and on series and parallel circuits.
Lesson 6 Factors Affecting Resistance: Why Long Thin Extension Cables Heat Up	We measured V and I for wires of different length, thickness and material, and for a wire that warmed up. We calculated $R = V/I$. We saw that longer wires and thinner wires gave a smaller current and so a higher R. Copper gave a low R, nichrome a higher R, and the bulb filament (30Ω in our demo) had a much higher R than copper connecting wire. A warm metal wire let less current flow.	Resistance increases with the length of a conductor, decreases when the cross-sectional area is larger (thicker wire), depends on the material, and for a metal increases as the temperature rises. $R = V/I$ (Lesson 5) lets us compare resistances. A higher resistance means less current for the same p.d. and more energy transferred as heat in the conductor.	A long thin extension cable has a high resistance, so when current flows more energy is transferred as heat and the cable warms up, unlike a short thick copper wire. Each bulb filament is a resistor with its own resistance (30Ω in our demo), unlike copper connecting wires. This helps explain why the way bulbs are connected changes the current, but we still need to find out how resistances combine in series and in parallel (Lessons 7 and 8).
Lesson 7 Resistors in Series: Why Three Bulbs Share the Battery and Glow Dim	In the series circuit of three identical 30Ω bulbs on the 9 V pack, the bulbs glow dim, the ammeter reads 0.10 A at every point, the voltmeter reads 3.0 V across each bulb and 9.0 V across the battery, and when one bulb is removed all bulbs go out and the current is 0 A.	For resistors in series, $R(\text{total}) = R_1 + R_2 + R_3$ ($30 + 30 + 30 = 90 \Omega$). The current is the same everywhere in the loop ($I = 9/90 = 0.10$ A), and the battery p.d. is shared between the bulbs (3.0 V each, adding to 9.0 V). Each bulb transfers $P = IV = 0.30$ W.	The three series bulbs are dim because their resistances add up to 90Ω , which limits the current to 0.10 A and gives each bulb only 3.0 V and 0.30 W instead of the 2.7 W it gets at full 9 V. In a series loop, a failed or removed bulb opens the only path so $I = 0$ and everything goes out, as in an old string of Christmas lights. Why the classroom tubes and parallel bulbs behave differently is still to be investigated in Lesson 8.

Lesson 8 Resistors in Parallel: Why One Failed Tube Does Not Darken the Others	We connected three identical 9 V, 2.7 W (30 Ω) bulbs on separate branches of the 9 V pack. Each had 9.0 V across it and 0.30 A through it, the battery current was 0.90 A and all bulbs glowed brightly (2.7 W each). When one bulb was removed, the other two stayed bright at 9.0 V and 0.30 A and the battery current fell to 0.60 A. In the Nairobi classroom, each tube draws 0.15 A, the four draw 0.60 A, and with one blown the other three draw 0.45 A.	In a parallel circuit each branch has the same p.d. as the source, the branch currents add up to the total current, and the effective resistance is found from $1/R = 1/R_1 + 1/R_2 + 1/R_3$, which gives 10 Ω for three 30 Ω bulbs. The effective resistance is less than the smallest branch resistance. A break in one branch does not change the p.d. or current in the other branches.	The other tubes stay lit because each tube is on its own branch directly across the 240 V supply, so when one branch is broken the others still have the same voltage and current (0.15 A each). Parallel bulbs glow brighter than series bulbs because each gets the full 9.0 V and 0.30 A (2.7 W), compared with 3.0 V, 0.10 A and 0.30 W in series. In a series string such as old Christmas lights, one failed bulb opens the only loop, so the whole string goes dark.
Lesson 9 Combined Circuits and Cells: Reducing a Mixed Circuit and Powering Homes from Kisumu to Kericho	We built a mixed circuit with one 30 Ω bulb in series with two 30 Ω bulbs in parallel on the 9 V pack. The bulb in series was brightest; the battery current was 0.20 A, the p.d. across the series bulb was 6.0 V, and each parallel bulb had 3.0 V and 0.10 A. When one parallel bulb was removed the other stayed lit. One cell gave 1.5 V, three cells in series gave 4.5 V and six cells in series gave 9 V; two cells in parallel still gave 1.5 V.	To solve a mixed circuit, reduce the parallel group first ($1/R = 1/30 + 1/30$, so 15 Ω), then add the series resistance ($30 + 15 = 45 \Omega$), then use $I = V/R = 9/45 = 0.20$ A and work back to find the p.d. and current in each part. Cells in series add their e.m.f.; cells in parallel give the same e.m.f. but last longer. Achieng's 12 V solar home system in Kisumu uses a higher e.m.f. source with lights wired in parallel.	The blown tube in Nairobi does not affect the other tubes because each tube is on its own parallel branch across the 240 V supply, while three series bulbs share the 9 V (3.0 V each) and carry only 0.10 A so they glow dim. A mixed circuit is a combination of both: the series part takes a bigger share of the p.d. and carries the total current, while the parallel part splits the current. Choosing how cells and bulbs are connected decides the p.d. and current each bulb receives, which explains the differences in brightness.
Lesson 10 Electrical Energy and Power: Why Series Bulbs Glow Dim and What the Tubes Cost	We observed that the three series bulbs (3.0 V and 0.10 A each) glowed dim, and the three parallel bulbs (9.0 V and 0.30 A each) glowed bright. We saw that the Nairobi classroom tubes are all on 240 V and stay lit when one blows. Our data card showed that a 60 W bulb uses more energy than a 9 W LED in the same time.	We learned that power $P = IV$ (in watts, 1 W = 1 J/s) and energy $E = IVt = Pt$. We learned that 1 kWh = 1000 W for 1 hour = 3.6 million J, and that cost = kWh $\times$ KSh 25. We calculated 0.30 W for a series bulb, 2.7 W for a parallel bulb, 0.72 kWh = KSh 18 for four tubes in 5 h, and KSh 7.50 for a 60 W bulb against KSh 1.13 for a 9 W LED.	Series bulbs are dim because each gets only 3.0 V and 0.10 A, which gives 0.30 W, while parallel bulbs each get the full 9 V and 0.30 A, which gives 2.7 W (9 times more). The parallel circuit takes more total power from the battery (8.1 W against 0.90 W), so its energy is used faster, which is why Kipchirchir was partly right. The other tubes stay lit because each tube is on its own branch at 240 V, and the school pays for the energy by the kWh, which can be reduced by using LEDs.
Lesson 11 Domestic Wiring and Safety: Why the Fuse Blows When a Socket Is Overloaded	We observed that a kettle (12.5 A) and a heater (8.33 A) on one socket draw 20.8 A, more than a 13 A fuse allows; a thin fuse wire melted when the current became too large; the plug has a fuse on the live wire and an earth wire; an extension cable that is long and thin heats up.	Appliances in a home are in parallel, so their currents add. $I = P/V$. Too much current (overload or short circuit) heats the wires; a fuse or circuit breaker on the live wire cuts the current. Earthing gives fault current a safe path and protects people from shock.	The fuse blows because the total current of 20.8 A exceeds the 13 A rating, so the thin fuse wire heats and melts and breaks the circuit before the cable overheats. The other tubes in the Nairobi classroom stay lit when one blows because they are in parallel, but the whole parallel system needs a fuse or breaker and earthing for safety.

Lesson 12 Model Revision and Application: Why the Other Tubes Stay Lit and Why Series Bulbs Glow Dim	We observed that three identical bulbs on a 9 V battery pack glow dim in series ($I = 0.10$ A, 3.0 V each, 0.30 W each, total $R = 90\ \Omega$) and bright in parallel (9.0 V and 0.30 A per branch, 2.7 W each, 0.90 A from the battery, effective $R = 10\ \Omega$). Removing one bulb in series puts all out, but in parallel the others stay lit, with the total current falling to 0.60 A. In the Nairobi classroom, when one of 4 tubes blows, the other three stay lit.	We learned that current is the flow of charge measured in series, that p.d. is energy per unit charge measured in parallel across a component, that $V = IR$, that resistances add in series ($R = R_1 + R_2 + R_3$) and combine in parallel ($1/R = 1/R_1 + 1/R_2 + 1/R_3$), that $P = IV$ and $E = Pt$ (so 144 W for 5 h = 0.72 kWh = KSh 18), and that fuses or breakers on the live wire and earthing protect people and wiring.	The other tubes stay lit because each tube is on its own branch directly across the 240 V supply, so if one branch breaks the others still have the same p.d. and current (0.15 A each; the total falls from 0.60 A to 0.45 A). Series bulbs are dim because they share the 9 V (3.0 V each) in one loop of $90\ \Omega$, so the current is only 0.10 A and the power only 0.30 W, whereas parallel bulbs each get 9 V, 0.30 A and 2.7 W. In series a failed bulb opens the one loop, like the old Christmas-light string. A classroom should therefore be wired in parallel with a fuse or breaker on the live wire and earthing.
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